GMR Institute of Technology
An Autonomous Institute Affiliated to JNTU-GV | GMRIT Training Tomorrow's Engineers Today

Temperature Based Ac Fan Control for Home Appliances

A Term paper report submitted in partial fulfilment of the requirement for the award of degree of

BACHELOR OF TECHNOLOGY

In

ELECTRONICS AND COMMUNICATION ENGINEERING

Submitted by

GANTA BHASKARA SRI SATYA BALAJ(23341A0446)

Under the esteemed guidance of

Dr. GEETAMMA TUMMALAPALLI

Associate Professor, Department of ECE,
GMRIT

GMR Institute of Technology

An Autonomous Institute Affiliated to JNTU-GV, Vizianagaram

(Accredited by NBA, NAAC with 'A' Grade & ISO 9001:2015 Certified Institution)

GMR Nagar, Rajam – 532 127, Andhra Pradesh, India
NOVEMBER 2025

GMR Institute of Technology
An Autonomous Institute Affiliated to JNTU-GV

GMRIT Training Tomorrow's Engineers Today

Department of Electronics and Communication Engineering

CERTIFICATE

This is to certify that the Term Paper entitled Temperature Based Ac Fan Control for Home Appliances submitted by **GANTA BHASKARA SRI SATYA BALAJI** (23341A0446), has been carried out in partial fulfilment of the requirement for the award of degree of Bachelor of Technology in Electronics and Communication Engineering of GMR Institute of Technology, Rajam affiliated to JNTU-GV, Vizianagarm is a record of bonafide work carried out by him under my guidance & supervision. The results embodied in this report have not been submitted to any other University or Institute for the award of any degree.

Signature of Supervisor
Dr. GEETAMMA TUMMALAPALLI
Associate Professor,
Department of ECE
GMRIT, Rajam.

Signature of HOD
Dr.V. Jagan Naveen
Professor & Head
Department of ECE
GMRIT, Rajam.

ACKNOWLEDGEMENT

It gives us an immense pleasure to express deep sense of gratitude to my guide, **Dr. GEETAMMA TUMMALAPALLI,** Associate Professor, Department of Electronics and Communication Engineering of wholehearted and invaluable guidance throughout the Term paper work. Without her sustained and sincere effort, this Paper work would not have taken this shape. She encouraged and helped us to overcome various difficulties that we have faced at various stages of our term paper work.

We would like to sincerely thank our Head of the Department, **Dr. V. Jagan Naveen**, for providing all the necessary facilities that led to the successful completion of our Term paper work.

I would like to sincerely thank our Associate Dean Academics, **Dr. M. V. Nageswara Rao**, for extending his support with all the necessary facilities that led to the successful completion of our Term paper work.

I would like to take this opportunity to thank our beloved Principal, **Dr. C. L. V. R. S. V Prasad**, for providing a great support to us in completing our Term paper and for giving us the opportunity of doing the term paper work.

I would like to take this opportunity to thank our beloved Director Education, **Dr. J. Girish**, for providing all the necessary facilities and a great support to us in completing the Term paper work.

I would like to thank all the faculty members and the non-teaching staff of the Department of Electronics and Communication Engineering for their direct or indirect support for helping us in completion of this Term paper work.

Finally, I would like to thank all of our friends and family members for their continuous help and encouragement.

GANTA BHASKARA SRI SATYA BALAJI (23341A0446)

ABSTRACT

This paper presents a simple and affordable solution to improve comfort and save energy in household ceiling fans. Many people in South India continue to use ceiling fans instead of air conditioners due to their lower cost and energy usage, but traditional fans cannot adjust their speed automatically as the room temperature changes. To address this problem, I designed a temperature-based automatic fan speed control system using an Arduino microcontroller.

The system continuously monitors the surrounding temperature and humidity using a DHT11/DHT22 sensor. Based on these readings, the Arduino controls a servo motor that rotates the knob of a capacitive regulator to set the appropriate fan speed. When the room temperature rises, the fan automatically speeds up, and when it cools down, it slows to a comfortable level, ensuring steady airflow without the need for manual adjustment . Unlike resistive regulators, the use of a capacitive regulator minimizes heat loss and enhances energy efficiency. This paper offers a cost-effective and practical approach for achieving comfort and energy savings in everyday household applications.

Keywords: Arduino Uno, Capacitive regulator, Automatic speed control , Energy saving

TABLE OF CONTENTS

1.INTRODUCTION

In most homes across South India, ceiling fans continue to be the most common and dependable means of staying cool in the warm and humid climate. They remain an essential fixture in both urban and rural households due to their simplicity, affordability, and energy efficiency. Unlike air conditioners, which require a considerable amount of electricity and involve high installation and maintenance costs, ceiling fans provide a cost-effective solution for everyday cooling needs. Their ability to circulate air effectively with minimal power consumption makes them ideal for households seeking comfort without the burden of high energy bills.

Despite their widespread use and reliability, traditional ceiling fans possess one major limitation—they cannot automatically adjust their speed in response to changing environmental conditions. In most southern regions, the temperature and humidity vary significantly between evening and late night. Typically, the air is hot and stuffy in the early evening hours, but as the night progresses, the surrounding temperature gradually drops, reaching its lowest point after midnight. This natural cooling pattern often creates discomfort for users who rely on fixed-speed fans. Many people find themselves either waking up during the night to reduce the fan speed or sleeping uneasily under a fan that feels too strong once the temperature has fallen.

This situation not only disturbs sleep and reduces comfort but also leads to unnecessary energy consumption, as the fan continues to operate at a higher speed even when it is no longer required. Over time, this wastage contributes to increased electricity usage, which could otherwise be minimized through intelligent control. Addressing this everyday inconvenience provides an opportunity to enhance comfort while promoting energy efficiency in an affordable manner.

The present project aims to design and develop a smart controller that can automatically regulate the speed of a conventional ceiling fan according to changes in room temperature and humidity. The primary objective is to provide a simple, low-cost, and user-friendly solution that can be easily implemented in ordinary households, particularly those without air conditioning systems. Instead of replacing existing fans with expensive smart appliances, the project focuses on upgrading traditional fans with an intelligent control mechanism that responds dynamically to environmental conditions.

At the core of this system lies an Arduino microcontroller, which functions as the central processing unit or “brain” of the setup. A DHT sensor is connected to the Arduino to continuously monitor the ambient temperature and humidity of the room. Based on the sensor readings, the microcontroller automatically adjusts the speed of the fan. When the temperature is high and the air feels warm or humid, the fan operates at a higher speed to enhance air circulation and provide a cooling effect. As the night progresses and the temperature gradually decreases, the controller reduces the fan speed accordingly. The transition between different speeds is designed to be smooth and gradual, ensuring consistent comfort without abrupt changes in airflow.

This project seeks to demonstrate how simple electronic components and basic programming can be effectively combined to solve a common household problem. The system exemplifies how affordable technology can be used to improve quality of life and promote responsible energy usage. By reducing the need for manual intervention and minimizing unnecessary power consumption, the proposed system not only enhances user comfort but also contributes to sustainable living practices.

Moreover, the development of such a device aligns with the broader movement toward smart and energy-efficient home solutions. As energy costs continue to rise and environmental concerns gain importance, even small innovations that optimize electricity usage can have a meaningful cumulative impact. The smart fan controller represents a practical example of how automation and environmental sensing can be integrated into everyday appliances, making them more adaptive, responsive, and efficient.

In essence, this project is guided by the goal of combining comfort, convenience, and conservation. It emphasizes that meaningful innovation does not always require complex or costly technology; rather, it can emerge from a thoughtful application of simple principles. By intelligently adjusting the fan’s speed according to temperature and humidity variations, the proposed system brings an element of smart functionality to an ordinary household device. Through this work, I aim to illustrate how accessible technology can be leveraged to create more comfortable living environments while simultaneously promoting energy efficiency and sustainability in daily life.

2.LITERATURE SURVEY

1.1 Literature Survey 1:

Ragul et al. (2024) – Automatic Speed Control of Ceiling Fan using PWM and Optocoupler.

Ragul, S., Yuvaraj, S., Vijayabalan, V., and Venkatasamy, B. (2024) designed and implemented an automatic ceiling-fan speed controller using a Pulse Width Modulation (PWM) technique with an optocoupler interface to reduce energy consumption. Their system used a microcontroller to sense room conditions and automatically adjust fan speed by varying the PWM duty cycle. The inclusion of an optocoupler ensured electrical isolation between the control circuit and the high-voltage driver, improving safety and reliability. The authors reported that automatic control not only improved user comfort but also reduced unnecessary power usage compared to manual regulators. The main advantage of this design lies in its simplicity, cost-effectiveness, and safety, though it remains an open-loop system without precise feedback. This work serves as a base model showing that efficient, low-cost control of ceiling-fan speed can be achieved using PWM and isolation techniques.

1.2 Literature Survey 2:

Kumar et al. (2023) – Power Quality Improvement in BLDC Ceiling Fan Drives

Kumar et al. (2023) focused on developing a Brushless DC (BLDC) ceiling fan drive with improved power quality and higher efficiency. Their work introduced an inverter-based drive using PWM control to shape input current, achieving near-unity power factor and lower total harmonic distortion (THD). By comparing a BLDC fan to a conventional AC induction fan, they demonstrated significant energy savings and smoother operation. The authors used simulation and experimental results to confirm that PWM-based electronic control allows precise speed regulation while minimizing losses. The main contribution of this study is the combination of energy efficiency and power-quality enhancement, which makes BLDC fans highly suitable for smart, low-power appliances. This paper complements Ragul et al.'s work by showing that when PWM control is applied to BLDC motors, even greater energy savings and performance gains can be achieved, though at a higher cost due to advanced electronics.

1.3 Literature Survey 3:

Sharma et al. (2018) – Energy-Efficient Ceiling Fans using BLDC Motor Technology

Sharma et al. (2018) conducted an experimental study comparing traditional induction-motor ceiling fans with newly developed BLDC-motor fans controlled through PWM techniques. The research revealed that BLDC fans consumed almost 50% less energy while maintaining the same airflow and comfort levels. The system used PWM-based commutation to control the motor torque and speed efficiently, resulting in reduced losses and improved overall performance. Their design showed that modern electronic control and BLDC technology can transform ordinary ceiling fans into smart, energy-saving appliances. Although the paper did not emphasize isolation or retrofit options, it proved the effectiveness of PWM in fan applications. When compared with Ragul et al. (2024), Sharma's work represents the next stage of development, where energy savings are maximized by integrating PWM control directly into BLDC-based fan designs rather than using it on conventional AC fans.

1.4 Literature Survey 4:

Nair et al. (2022) – PWM-Based Single-Phase Fan Regulator with High Power Factor

Nair et al. (2022) presented a single-phase electronic regulator for ceiling fans that used PWM control instead of the conventional triac-based phase control. Their circuit design achieved smoother speed variation, lower power loss, and improved power factor compared to traditional regulators. The authors implemented filtering and isolation circuits to suppress electromagnetic interference (EMI) and enhance user safety. Experimental analysis showed that PWM control provided linear and efficient speed regulation with less heat generation in the circuit. The study concluded that PWM-based regulators are a better choice for domestic fan control, offering both efficiency and power-quality benefits. This paper closely relates to Ragul et al. (2024) since both focus on PWM-controlled AC fans, though Nair et al. further improved power quality and EMI performance, which could enhance Ragul's proposed design if combined.

1.5 Literature Survey 5:

Automatic Fan Speed Control System Using Microcontroller" (Bagal et al., 2021)

Bagal, Chakane & Sachin (2021) developed an automatic fan speed control system that uses a temperature sensor (LM35DZ), a PIC16F877A microcontroller and a BLDC motor, with the aim of reducing manual effort and improving comfort. They built the circuit so that as the ambient temperature rises, the fan speed increases automatically, and as it cools down, the speed reduces. The system even includes an LCD to display temperature and speed for user feedback. The benefit is clear: less manual adjustment, more automation. On the flip side, their setup uses a BLDC motor (which is more complex than a standard AC ceiling fan) and is more of a prototype than a mass-home solution. This work is relevant to Ragul et al. because it shows how sorted electronics and PWM/motor drive can achieve automatic speed control; Ragul's work focuses on making that applicable in everyday ceiling fan contexts with optocoupler isolation.

1.6 Literature Survey 6:

Design and Implementation of BLDC Motor Controller for Energy Efficient Fan (Patel, S & Shukla, 2019)

Patel, Kishan & Shukla (2019) presented a low-cost controller for BLDC motors targeted at household fan applications. They used an STM32F407 microcontroller to generate PWM signals and drive a BLDC motor, emphasising energy efficiency for ceiling or pedestal fans. Their results showed that using such electronics makes the fan more efficient and offers better speed control compared with conventional systems. The strength here is clear: good control, good efficiency. The trade-off is that BLDC motors & their electronics are more expensive and need more design effort. For Ragul et al., this indicates the "high-efficiency" end of the spectrum: if you redesign the motor+controller entirely you can get big savings; Ragul's approach is more retrofit-friendly for existing motors but still uses PWM and isolation.

1.7 Literature Survey 7:

Design of Multisensor Automatic Fan Control System Using Sugeno Fuzzy Method (Willy Robson et al., 2024)

Robson, Ernawati & Nugrahaeni (2024) introduced an automatic fan-control scheme that uses multiple sensors (temperature + humidity) and a Sugeno-type fuzzy logic controller to decide when to turn the fan on/off and at what speed. Instead of only temperature thresholds, they incorporate humidity and potentially user comfort factors. The system can switch motor speed and also shut the fan when not needed. The main advantage: smarter decision logic, better comfort + potential energy saving. The limitation: more complex sensor setup and logic, potentially higher cost and complexity than simpler threshold systems. For Ragul et al., this paper suggests how one could extend their system further: after their basic automatic PWM+optocoupler design, one could add fuzzy logic and multi-sensor input to improve comfort and efficiency.

1.8 Literature Survey 8:

Design and Implementation of an Electronic Fan Regulator for Reduced Harmonics and Ripple Free Speed (Kiriella, 2021)

Kiriella (2021) focused on an electronic fan regulator design (rather than full smart control) with the goal of reducing waveform distortion, current harmonics, speed ripple, and motor noise when controlling ceiling fans. Their design replaces old resistive or simple inductive regulators with an electronic regulator that chops the waveform (essentially an advanced PWM/triac switch) and includes filtering to reduce harmonics and maintain better power factor. The benefit: smoother operation, less motor stress and better power-quality. The downside: the design may not automatically sense temperature or adjust speed for comfort or energy saving per se—it focuses on improved regulator electronics. In the context of Ragul et al., this is a helpful complement: while Ragul uses PWM + optocoupler for speed control and energy saving, Kiriella shows that regulator design also matters for quality, noise, and harmonics in fan systems.

3.METHODOLOGY

System Overview

An Arduino Uno is the main controller. It reads the room temperature from a sensor and automatically adjusts the fan speed by moving a servo motor that turns the existing regulator knob.In place of Arduino Uno we can use other Microcontrollers like Arduino Nano,esp32.

Sensor Setup

A temperature sensor DHT22 is connected to the Arduino's analog input. This lets the microcontroller constantly take the room conditions and convert them into temperature and Humidity values in real time.

Servo connection

A small servo motor is mechanically fixed to the fan regulator. Different servo angles set for different fan speeds, so rotating the servo changes speed of fan just like a person turning the knob and adjusting the speed.

Control Logic

The Arduino program continuously reads the temperature. Depending on the reading, it chooses a certain servo angle: 1) Cool - low fan speed

2) Warm - medium fan speed

3) Hot - high fan speed

This logic will update regular interval of time and detects changing temperature.

Software Implementation

The sketch was written in the Arduino IDE using the Servo library. The code maps temperature readings to servo angles, and the Serial Monitor was used during testing to find thresholds values and respective angles.

Testing

During testing, I observed the temperature readings and adjusted the servo positions so that each angle matched a comfortable fan speed on our regulator. This make smooth, automatic operation throughout the night.If we need constant speed, we can turn off the device and can operate manually.

Outcome

The fan slows down automatically as room cools for below threshold level of temperature , saving energy and keeping airflow comfortable without the user needing to wake up at night and to adjust it.

Modification

Original System

Fan speed changed electronically by the microcontroller, but limited to ON/OFF or two-step speeds.

Modified System

Replaced the simple electronic speed control with a servo motor coupled to a capacitive regulator. The servo rotates the knob of a capacitive regulator in real time according to the temperature readings. This physically adjusts the fan's capacitor-based speed control, just like a manual regulator but automated.

Capacitive Regulator

A capacitive regulator is a highly efficient method for controlling the speed of ceiling fans. Unlike resistive regulators, which reduce speed by wasting energy in the form of heat, a capacitive regulator uses the principle of phase shift which was created by a capacitor.

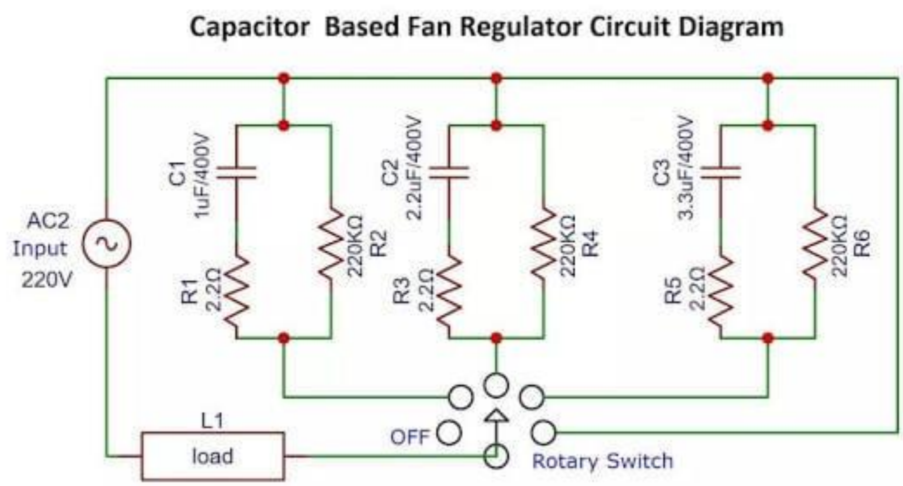


Fig 1: Capacitive Regulator

Fig 1 Shows When a capacitor is connected in series with the fan motor, it changes the phase difference between voltage and current. This effective voltage and current supplied to the fan motor, which in turn controls its speed. By switch to different capacitor values they have different phase shifts, then the fan can operate at multiple speed levels low, medium, or high without wasting power because capacitor is an active element and stores energy and very less loss of power will be there.

Energy Efficient:

very low power lost as due to heat.

Smooth Operation:

Provides stable speed control without sudden voltage drops.

Durability:

Simple design with fewer losses it has long life and reliable performance.

Servo connections

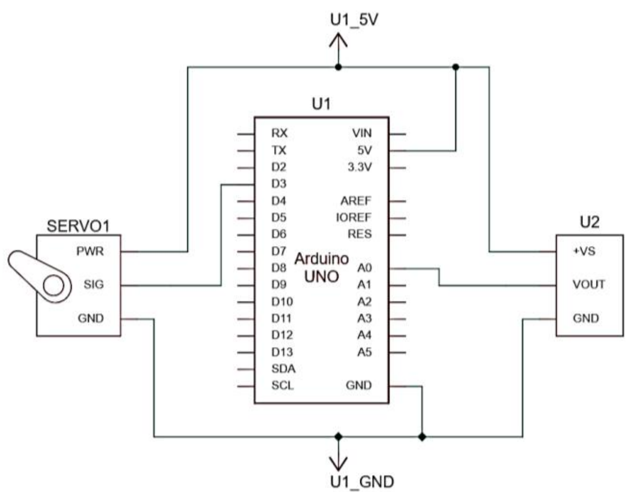
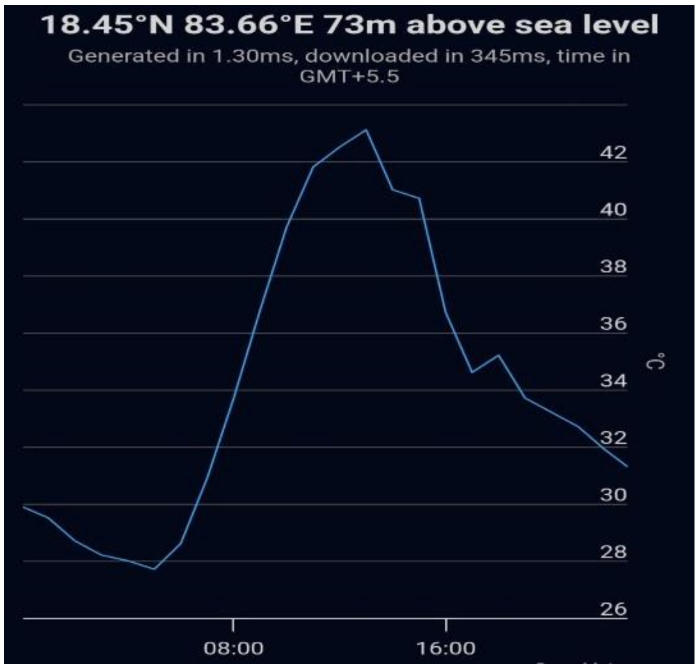


Fig 2 : Microcontroller connections

Fig 2 shows a simple control setup where an Arduino UNO (U1) drives a servo motor (SERVO1) using a separate power supply module . The Arduino receives 5 V from the main supply (U1_5V) and generates a control signal to control the servo motor angle. The servo's power pin is connected directly to regulated output of U2 to provide enough current without making the excess current for Arduino, while all grounds from the Arduino, the servo, and the power module are tied together (U1_GND) to make as common reference.

This arrangement allows the Arduino to send accurate PWM pulses to the servo for smooth and correct position control while maintaining stable power for both devices.

Temperature graph



Graph 1: Temperature variations

Graph 1 represents a normal day's temperature variation in a South Indian location mainly in summer time The graph shows how the climate here is often unpredictable and irregular. Early morning temperatures is around 29–30 °C, then steadily rise to a peak about 41–42 °C by midday. After this high point the temperature drops slowly in the late afternoon and continues to cool during the night. In many parts of South India, the temperature decreases even more rapidly after midnight, with noticeably cool conditions developing after 12 a.m., making the day–night temperature swing between hot and cool.

4.RESULTS

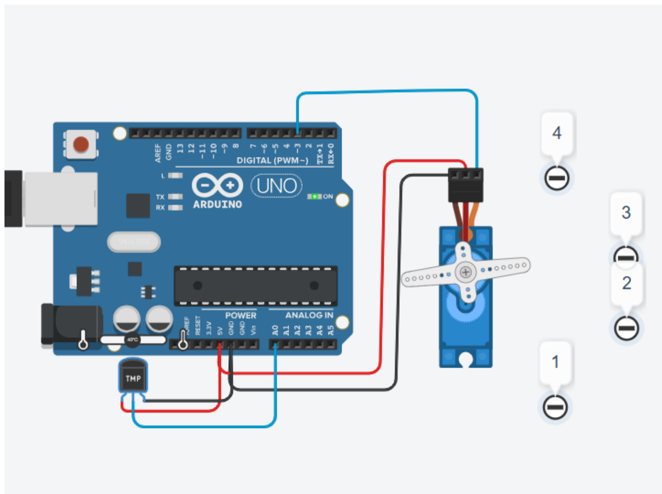


Fig 3 : Interfacinng with Arduino

The system responded well to temperature changes. As the room warmed up, the servo motor smoothly turned the fan regulator to higher levels as shown in **Fig 3**, and when the room cooled down, it reduced the speed automatically. The fan ran slower at lower temperatures and faster when it was hot, maintaining a comfortable airflow without any sudden changes.

At temperatures below 25 °C, the fan speed stayed around 200 rpm, giving a gentle breeze. Between 25 °C and 35 °C, it increased to about 280 rpm, and between 35 °C and 45 °C it reached around 370 rpm. Above 45 °C, the fan ran close to 420 rpm, providing strong air circulation.

Because the speed adjusted automatically, there was no need to change it manually during the night. This also helped in saving around 30–40% power, since the fan used less energy when running at lower speeds. The capacitive regulator further improved efficiency by reducing heat losses.
Overall, the system maintained comfort, worked smoothly, and reduced energy consumption, making it an effective and practical solution for everyday use.

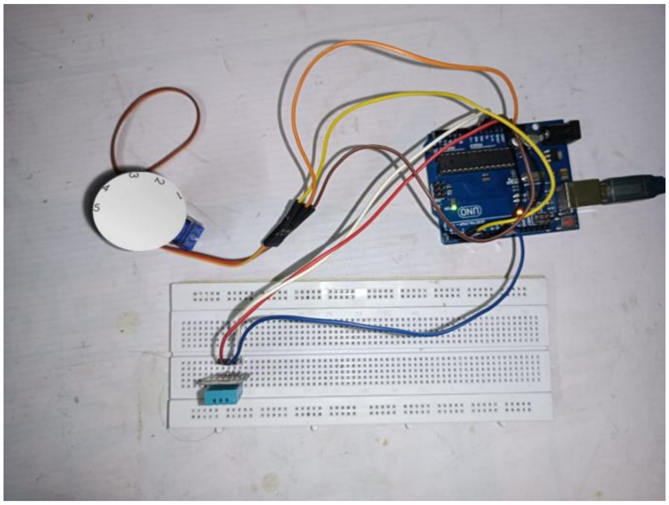
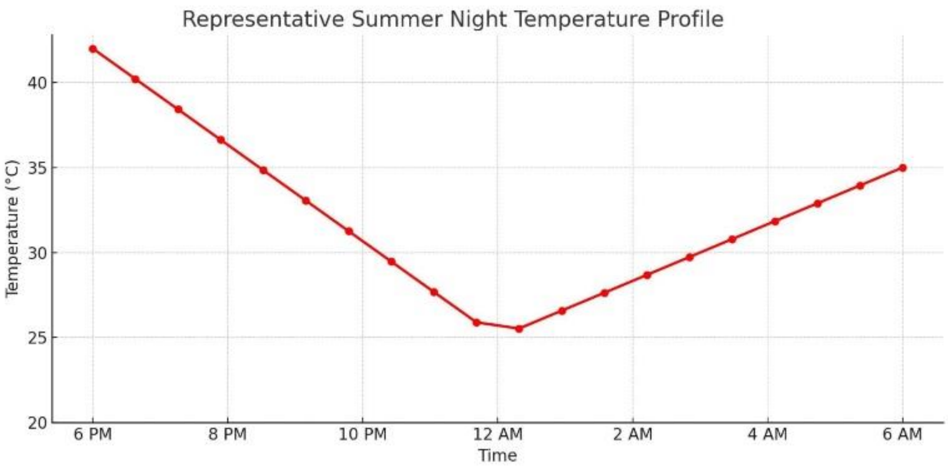


Fig 4 : Practical Connections

Table 1:Temperature and respective speed

Level	Temperature	Speed(rpm)
1	<25	200
2	25-35	280
3	35-45	370
4	>45	420+

Here the table represents 4 levels of the Capacitive Regulator . Based on the Temperature the level was changed by the servo motor and each level has specific rpm (Revolutions per minute).Lower rpm means the fan will run slow and low air with power saving .Higher rpm make higher breeze. Other two levels are for moderate wind.



Graph 2 : Summer Night temperature profile

5.CONCLUSION

This work shows that a ceiling fan can automatically maintain a comfortable airflow by responding to changes in room temperature. The system continuously reads the temperature and adjusts the fan speed smoothly without the need for any manual control. By using a servo motor connected to a capacitive regulator, it was able to turn the fan knob gradually, just like a person would do, but with better accuracy and consistency.

The results clearly indicate that automatic control helps in saving energy while keeping the comfort level steady. When the temperature drops, the fan slows down, and when it rises, the fan speeds up again. This balance ensures that electricity is not wasted at cooler times, reducing overall power consumption by nearly 30–40%. The use of a capacitive regulator further improves performance by minimizing heat loss, making the operation more efficient compared to traditional resistive regulators.

Another advantage of this design is that it can be used in any normal household without the need for complicated wiring or expensive components. The system's simplicity and low cost make it suitable for homes where air conditioning is not commonly used but comfort is still important. It also offers the convenience of both automatic and manual operation, giving users full flexibility.

In conclusion, automatic temperature-based fan speed control proves to be a practical and energy-saving approach for everyday living. It provides a comfortable environment, operates quietly, and reduces power bills with minimal effort. Such smart control systems can play an important role in future energy efficient homes, especially in warm regions like South India where temperature varies greatly between day and night.

6.REFERENCES

[1] S. Ragul, S. Yuvaraj, V. Vijayabalan, and B. Venkatasamy, “Design and implementation of automatic speed control of ceiling fan through PWM technique with optocoupler to reduce energy consumption,” in Proc. 3rd Int. Conf. Comput. Sci. Softw. Eng. (IC3SE), 2024. IEEE. doi: 10.1109/IC3SE62002.2024.10593059.

[2] S. Ragul, S. Yuvaraj, V. Vijayabalan, and B. Venkatasamy, “IoT based smart controller for ceiling fan,” in Proc. 7th Int. Conf. Trends Electron. Informat. (ICOEI), 2023, pp. xxx–xxx. IEEE. doi: 10.1109/ICOEI56765.2023.10125708.

[3] S. Das, S. Pal, T. Mukhopadhyay, and S. Nath, “Automatic temperature controlled fan,” Int. J. Recent Res. Electr. Electron. Eng., vol. 11, no. 3, 2024. doi: 10.5281/zenodo.13353749.

[4] Tinkercad, “Brilliant Kup Wluff circuit design,” Tinkercad, n.d. [Online]. Available: https://www.tinkercad.com/things/2HP8O5T4YkD-brilliant-kup-wluff/editel?returnTo=https://www.tinkercad.com/dashboard/designs/circuits.

[5] Stack Exchange, “Fan speed control using relays,” Electronics Stack Exchange, 2022. [Online]. Available: https://electronics.stackexchange.com/questions/637693/fan-speed-control-using-relays.

[6] Open-Meteo, “Historical weather API,” Open-Meteo, n.d. [Online]. Available: https://open-meteo.com/en/docs/historical-weather-api.

[7] A. K. Patil, “Design and Implementation of Arduino Based Temperature Sensing Device,” Int. J. Res. Publ. Rev. (IJRPR), vol. 3, no. 12, pp. 1407–1409, Dec. 2022.

International Journal of Research Publication and Reviews

(Open Access, Peer Reviewed, International Journal)

(A+ Grade, Impact Factor 6.844)

ISSN 2582-7421

Sr. No: IJRPR 137256-1

Certificate of Acceptance & Publication

This certificate is awarded to "Ganta Bhaskara Sri Satya Balaji", and certifies the acceptance for publication of paper entitled "Temperature Based Ac Fan Control for Home Appliances" in "International Journal of Research Publication and Reviews", Volume 6, Issue 10 .

Signed Aashish Agarwal

Editor-in-Chief
International Journal of Research Publication and Reviews

Date 05-10-2025